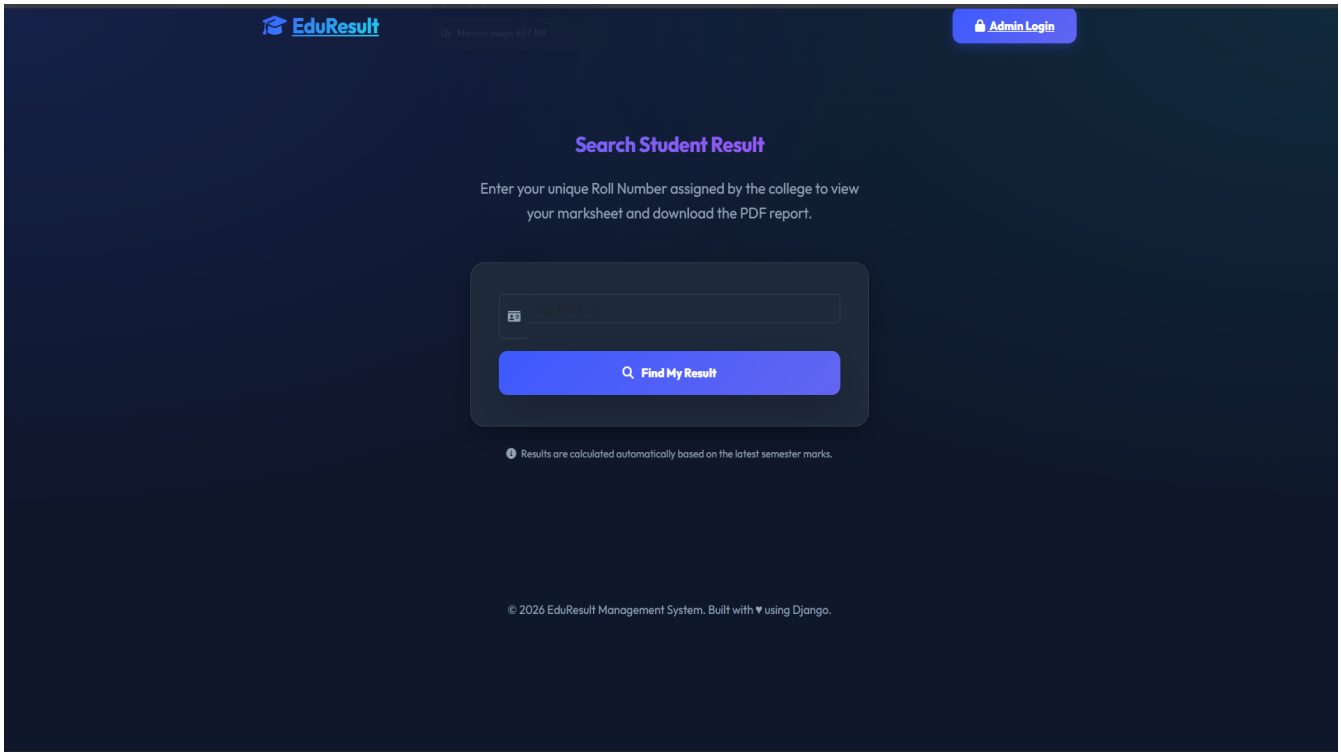
Student Result Management System

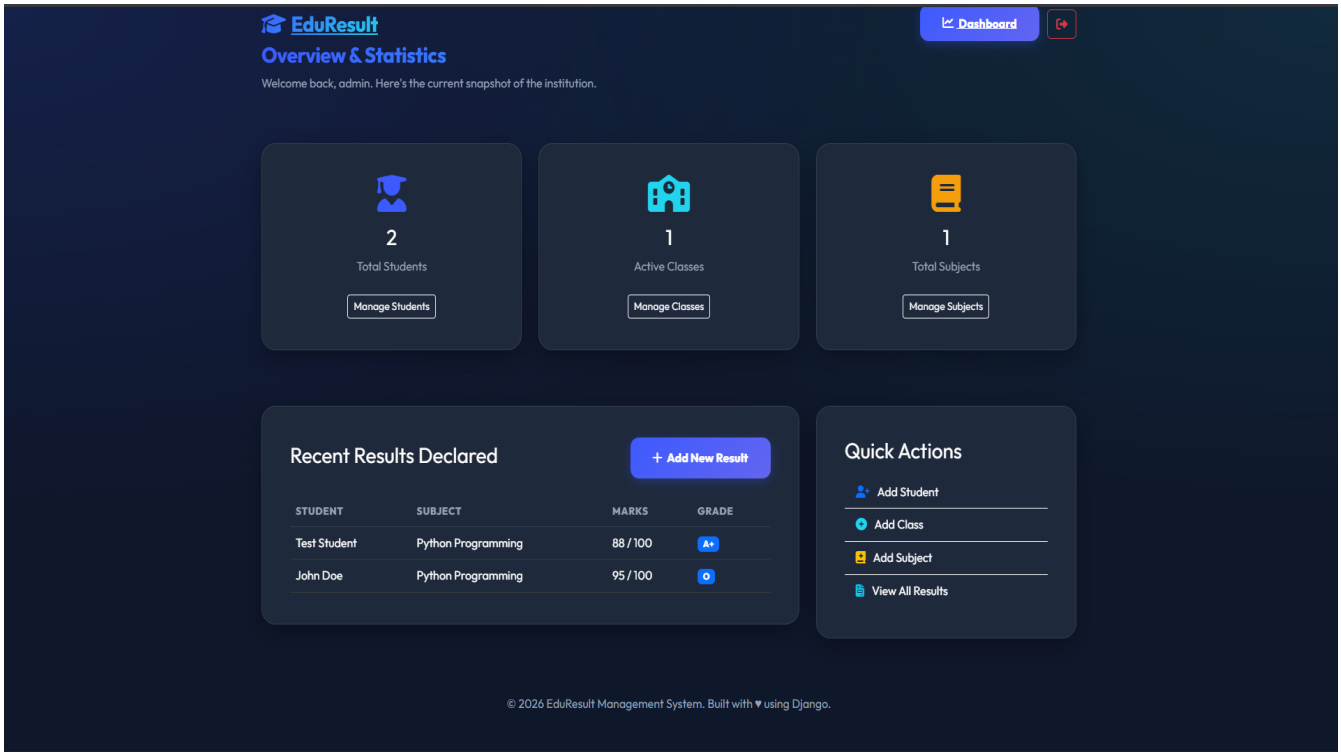
1. Requirement & Data Modeling

Class, Subject, Student, Result entities form the database core. Result links student and subject with marks.



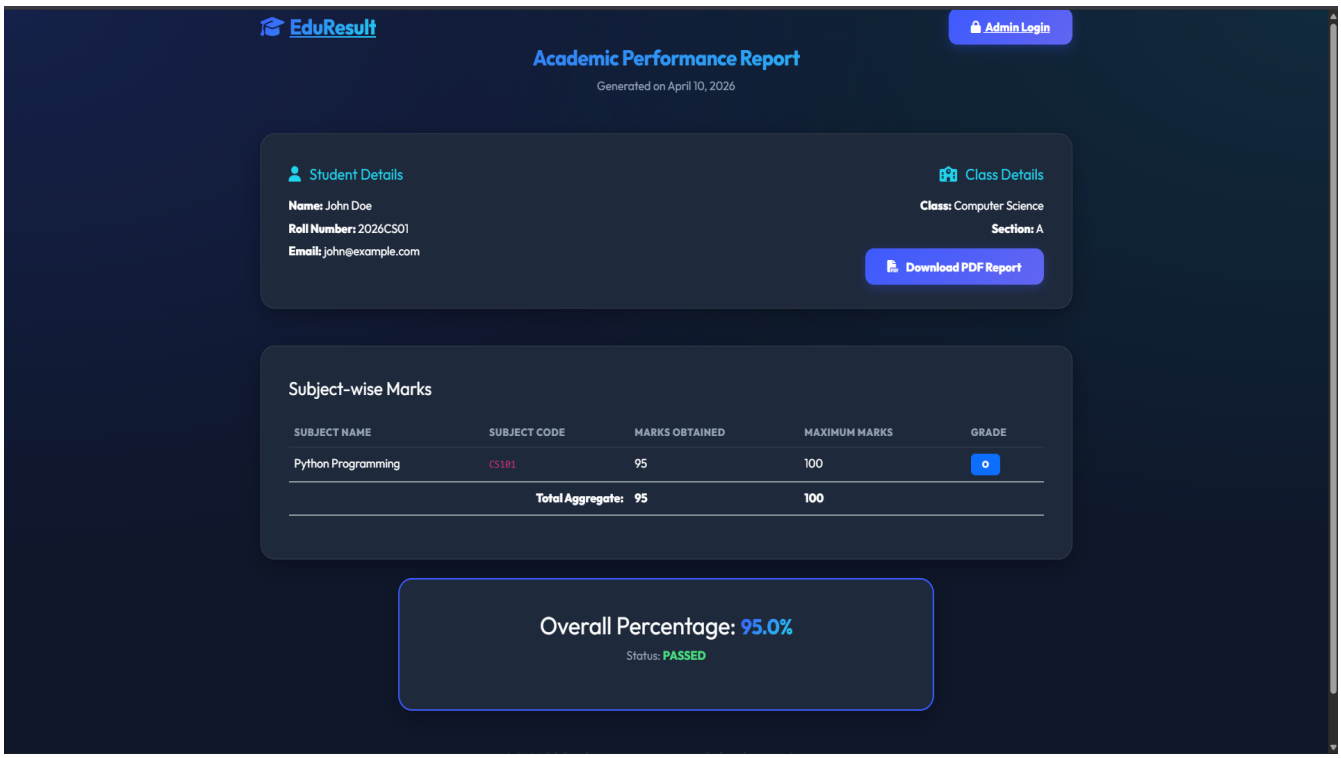
2. Backend Logic

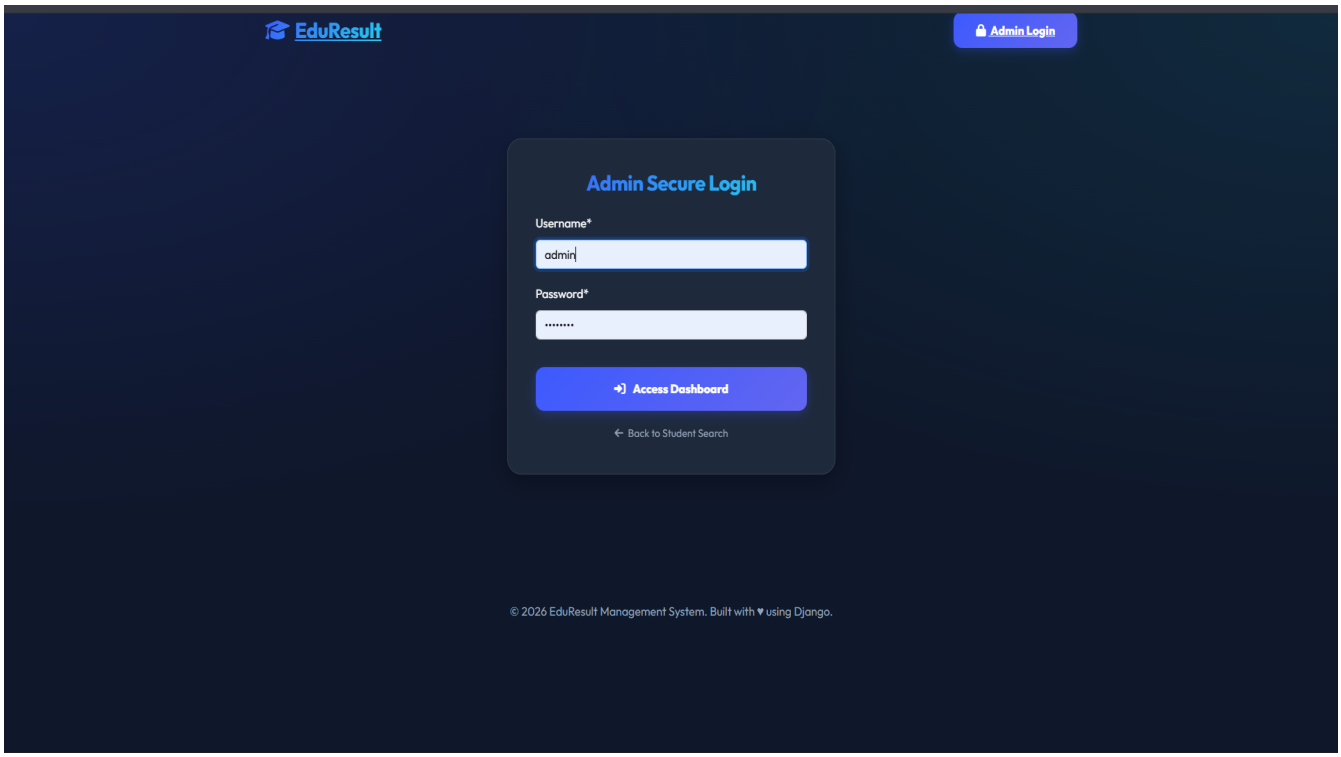
Authentication using LoginRequiredMixin. Grade calculation logic (O, A+, A, etc). CRUD operations using Django CBVs.



3. Student Discovery Workflow

Student enters roll number. System queries database. Marks aggregated and percentage calculated. Result displayed instantly.



4. PDF Generation Lifecycle

HTML template -> xhtml2pdf conversion. Generated PDF returned via HttpResponse.